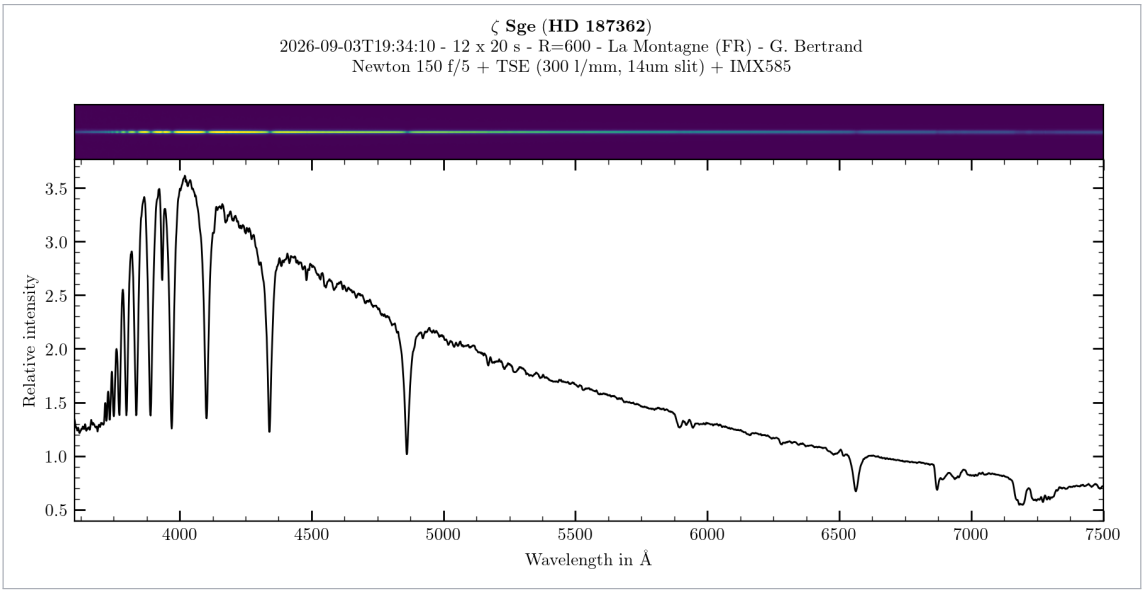
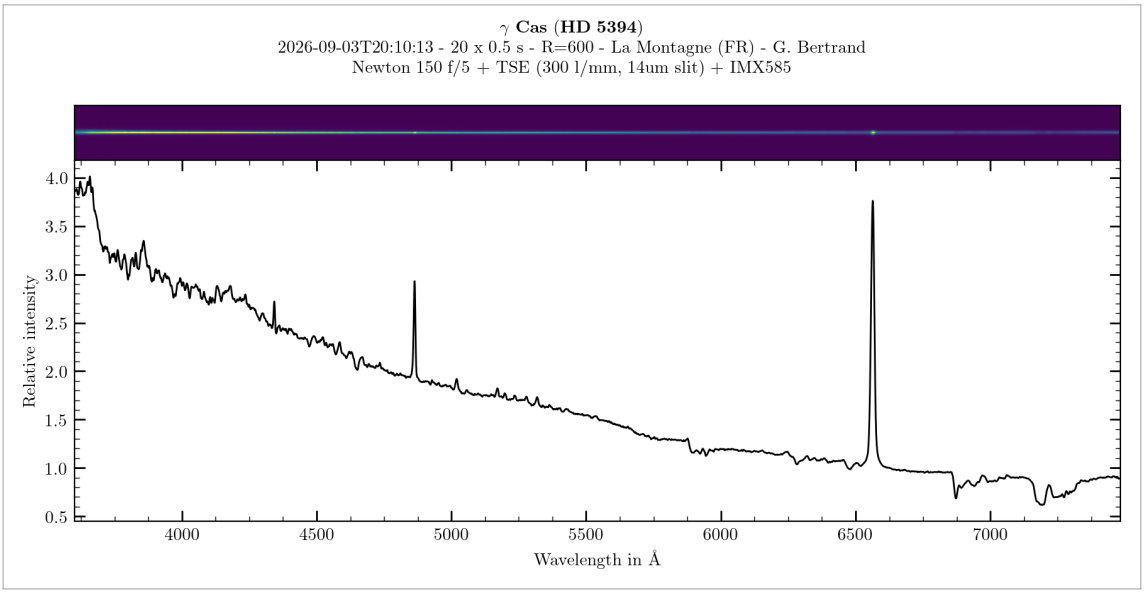
TinyStar’Ex

Exemples de spectres

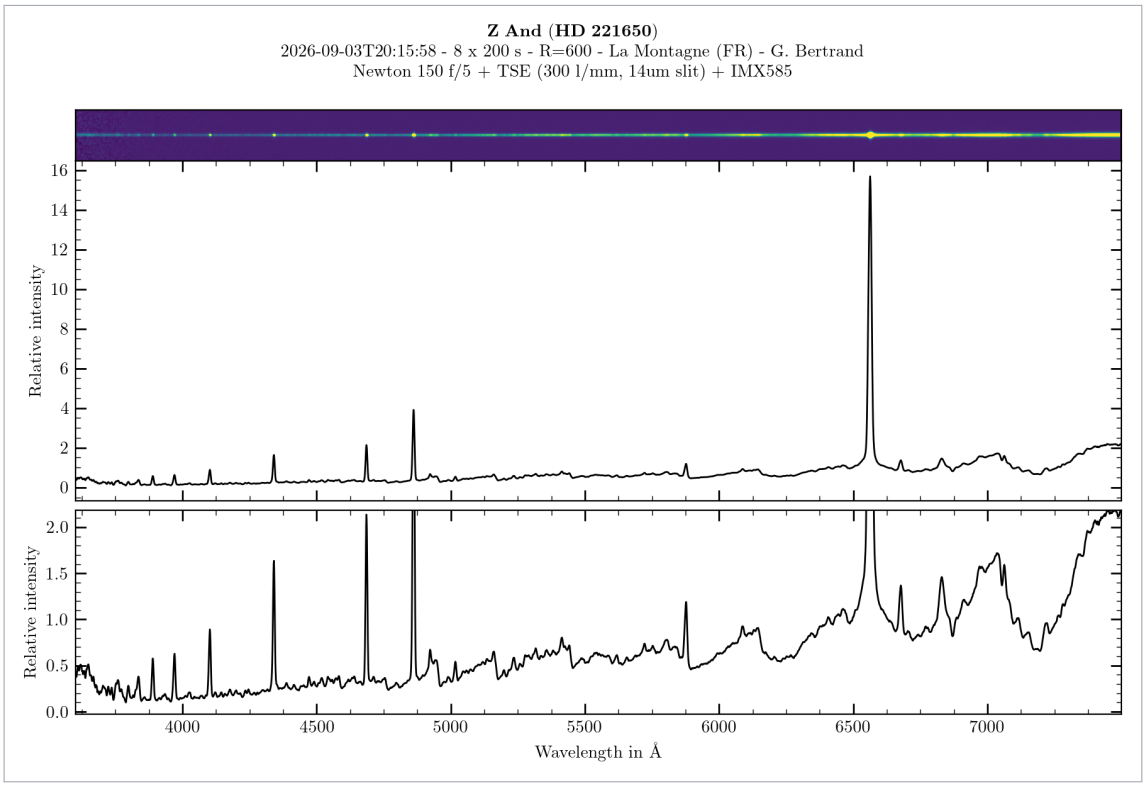
Spectre de l’étoile ζ Sge. TSE sur télescope Newton de 150 mm f/5. Guillaume Bertrand.



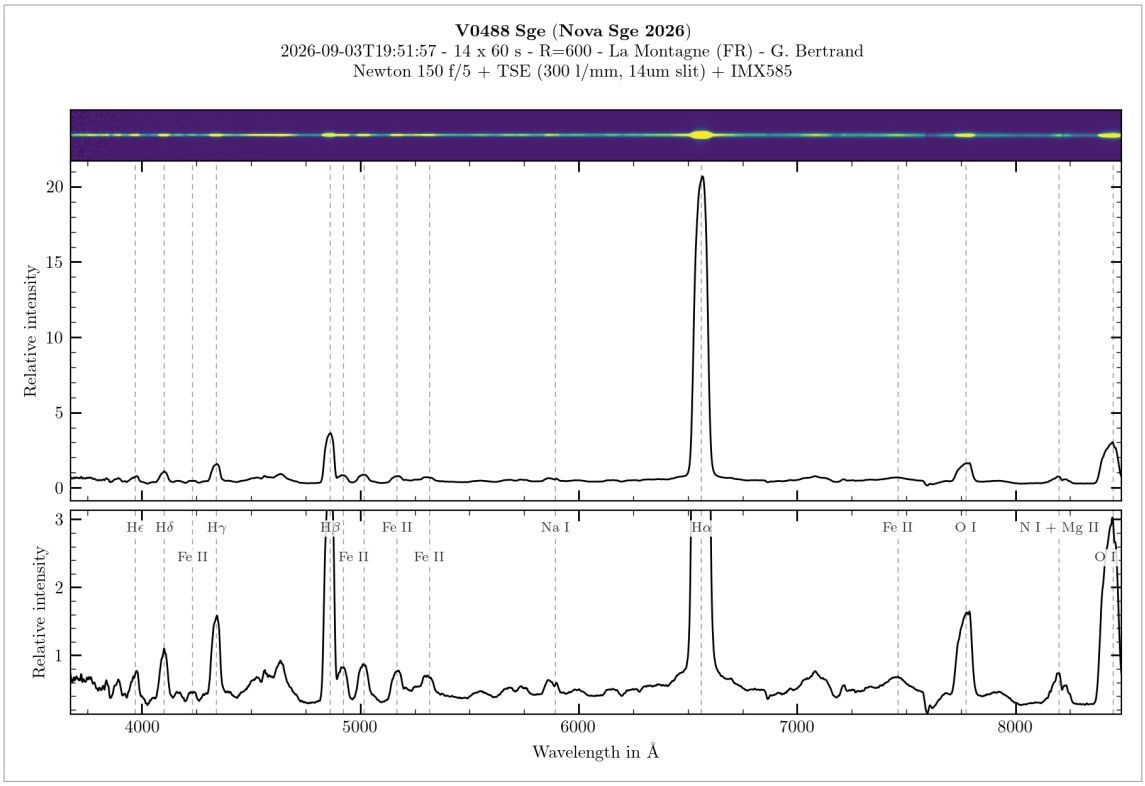
Spectre de l’étoile Be γ Cas. TSE sur télescope Newton de 150 mm f/5. Guillaume Bertrand.



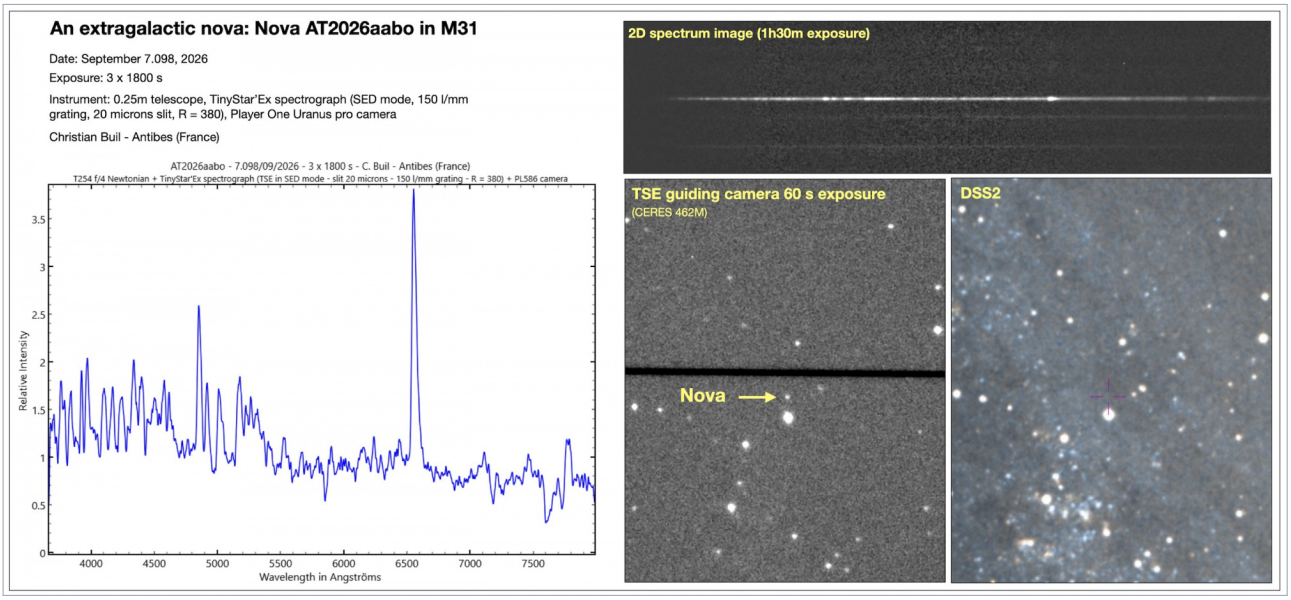
Spectre de l'étoile symbiotique γ Cas. TSE sur télescope Newton de 150 mm f/5. Guillaume Bertrand.



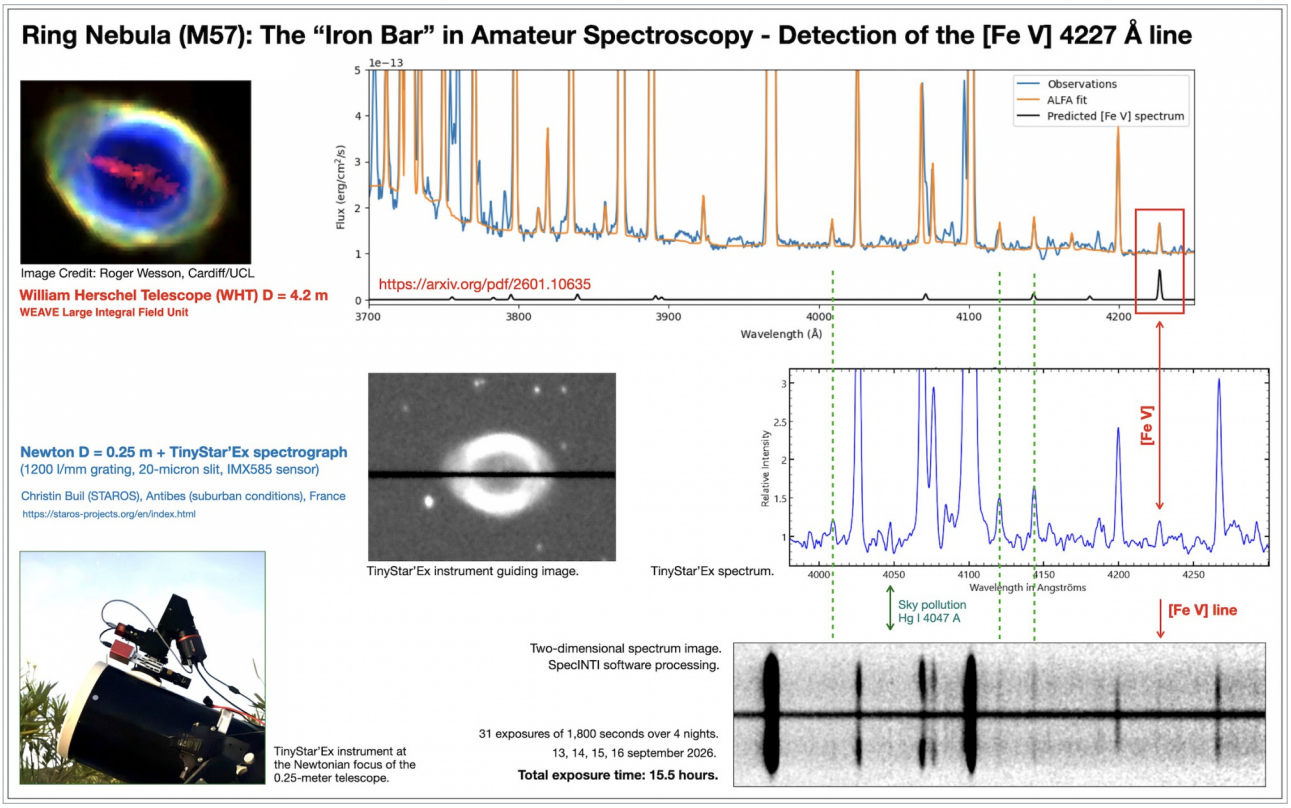
Spectre de la nova V488 Sge. TSE sur télescope Newton de 150 mm f/5. Guillaume Bertrand.



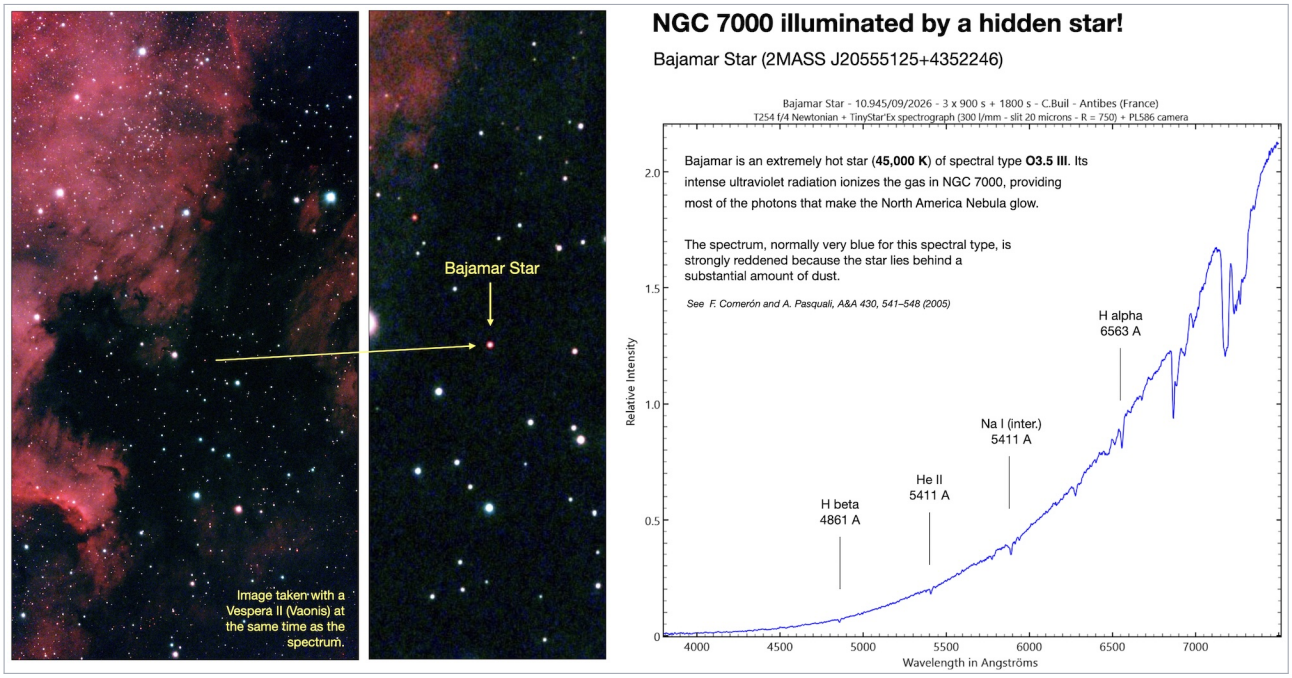
Spectre de la nova extragalactique AT2026aabo (dans Messier 31). TSE sur télescope Newton de 250 mm f/4. Christian Buil.



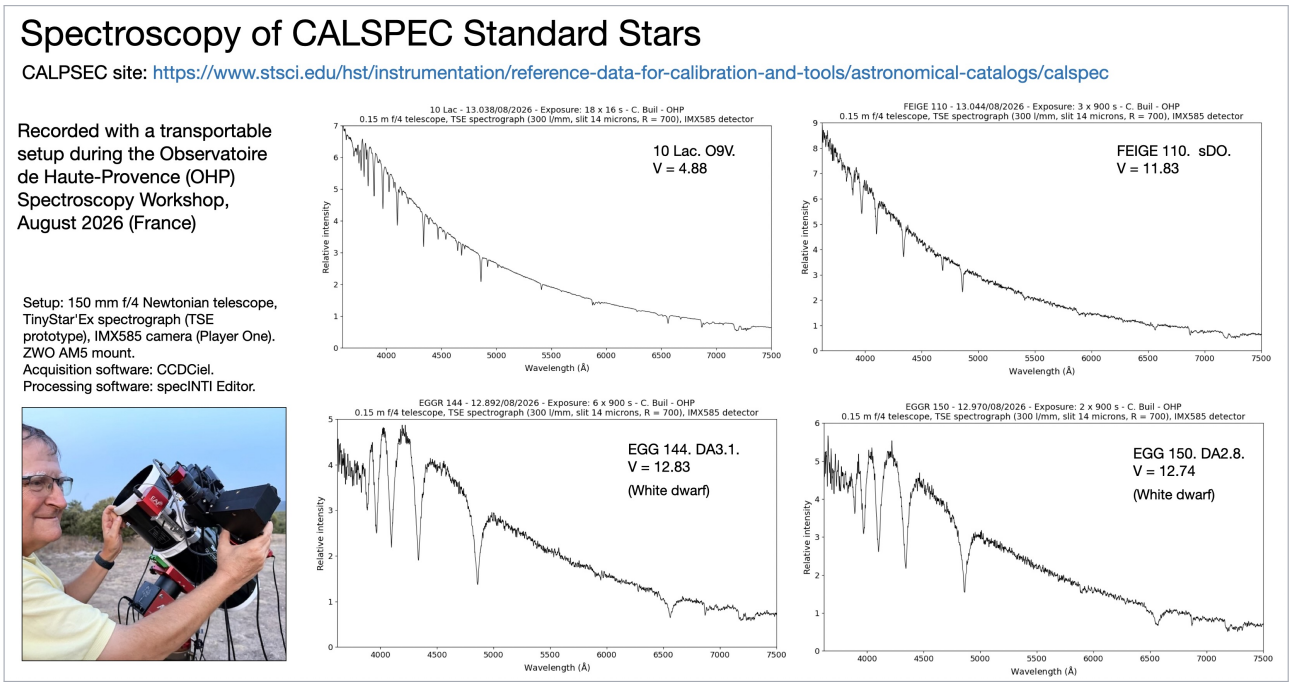
Spectre de la barre métallique récemment découverte dans Messier 57 (pose longue !). TSE sur télescope Newton de 250 mm f/4.



Spectre de l'étoile cachée qui illumine la grand nébuleuse NGC 7000 (America). TSE sur télescope Newton de 250 mm f/4.

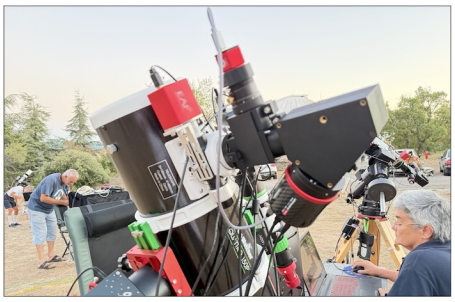


Spectres de standards spectrophométriques (étoiles très chaudes et naines blanches). TSE sur télescope Newton de 150 mm f/4.

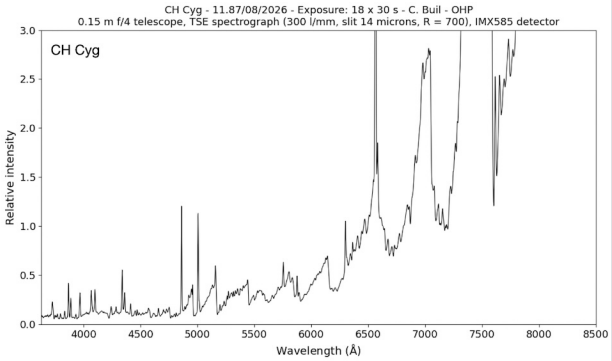


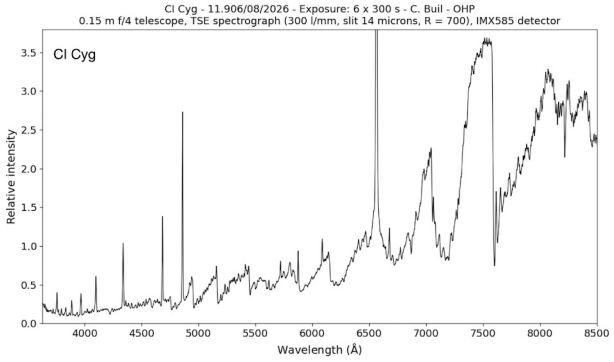
Spectres d’étoiles symbiotiques avec un télescope très compact. TSE sur télescope Newton de 150 mm f/4.

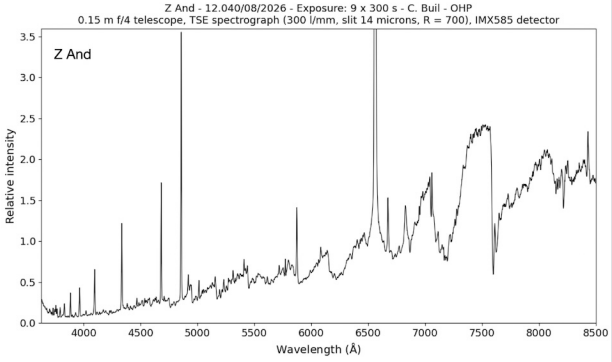
Observation of symbiotic stars during the 2026 spectroscopy workshop at the Observatoire de Haute-Provence (France). STAROS association



The D=150 mm f/4 telescope equipped with a TinyStar'Ex spectrograph. The equatorial is a ZWO AM5. Acquisition: CCDCiel. Processing: specINTI Editor.



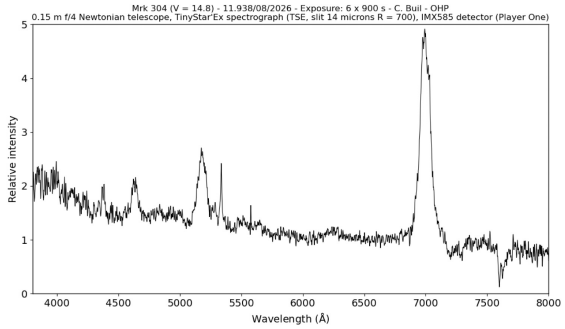




Spectre de la galaxie Markarian 304 selon le diamètre du télescope et le site d’observation (ciel pollué ou non par la lumière des villes)..

Observation of the galaxy Markarian 304 (Seyfert type 1 galaxy, V = 14.7) - Comparison between a professional observatory site and a suburban site

Note: the spectrograph and exposure time are the same, but not the telescope (0.150 m vs 0.254 m)





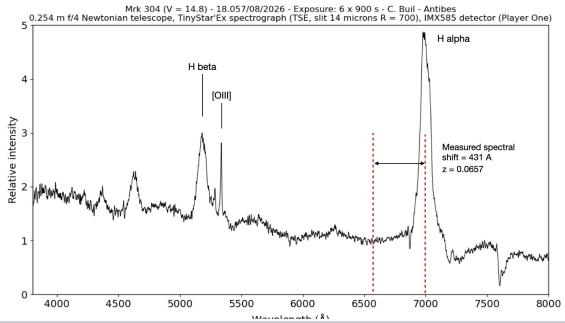
Observatoire de Haute-Provence (OHP)

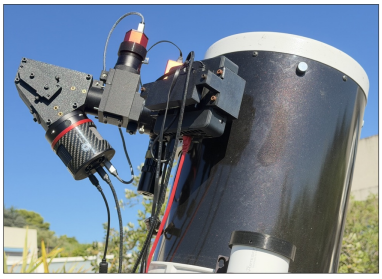
OHP 2026 Spectroscopy Workshop (STAROS)

0.15 m f/4 Newtonian telescope - Ti**nyStar'Ex** (TSE) spectrograph - IMX585 detector.

14-micron slit (4.8 arcseconds on sky).
Exposure: 6 x 900 seconds

Resolving power at 6563 Å (R): **700**
Signal measured at 6563 Å: **3600 ADU**
Measured signal-to-noise ratio: **17**





Observatoire Antibes Saint-Jean

Suburban conditions

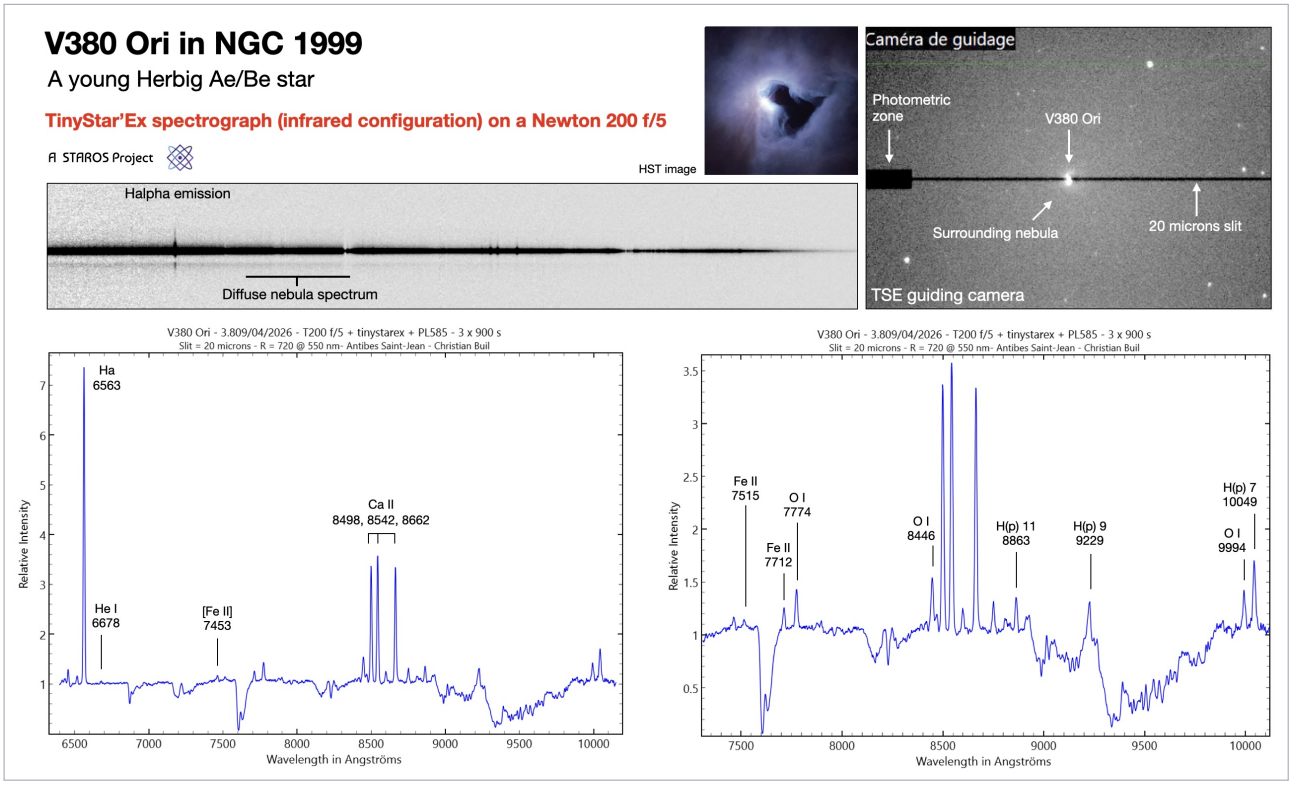
0.254 m f/4 Newtonian telescope - Ti**nyStar'Ex** (TSE) spectrograph - IMX585 detector.

14-micron slit (2.8 arcseconds on sky).
Exposure: 6 x 900 seconds

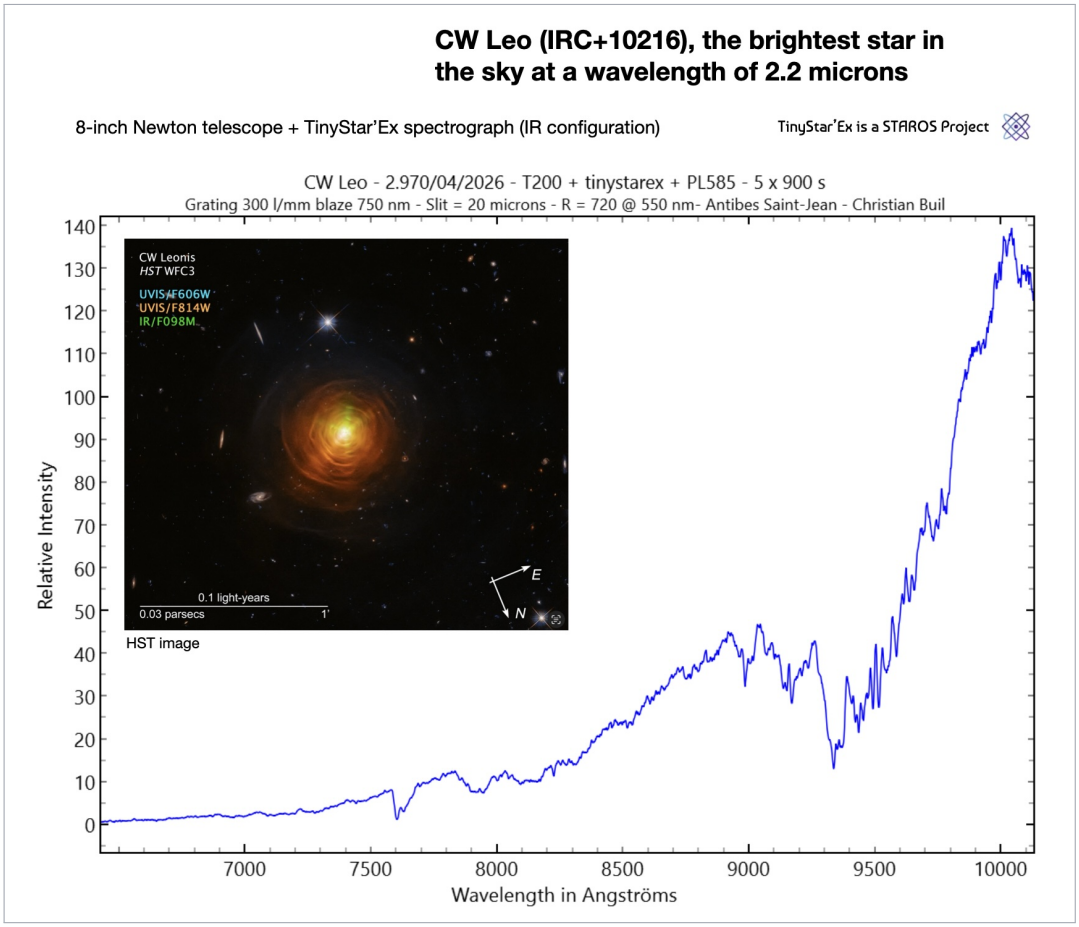
Resolving power at 6563 Å (R): **700**
Signal measured at 6563 Å: **13500 ADU**
Measured signal-to-noise ratio: **36**

G = 0.895 e-/ADU, RON = 1.02 e-, Dark = 0.0019 e-/s

Spectre de l'étoile jeune V380 Ori. Il illustre les excellentes performances de TinyStar'Ex dans le proche infrarouge : l'optique donne une image nette jusqu'à 1 micron de longueur d'onde, et même au-delà.



CW Leo : la plus brillante étoile du ciel…dans l'infrarouge !



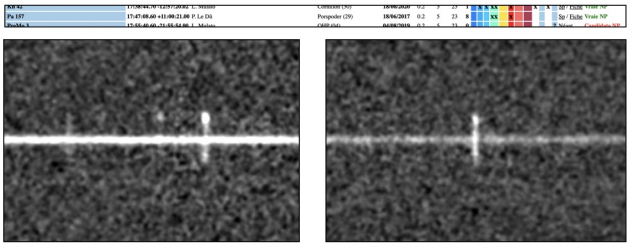
Observation en ville du spectre d'une nébuleuse ultra-faible. Dans ces conditions, cet objet resterait invisible sans spectrographe.

TinyStar'Ex

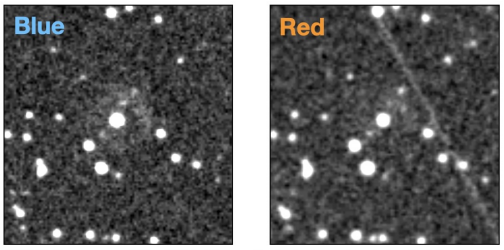
Pa 157: an ultra-faint planetary nebula

Amateur astronomers regularly discover new planetary nebulae.

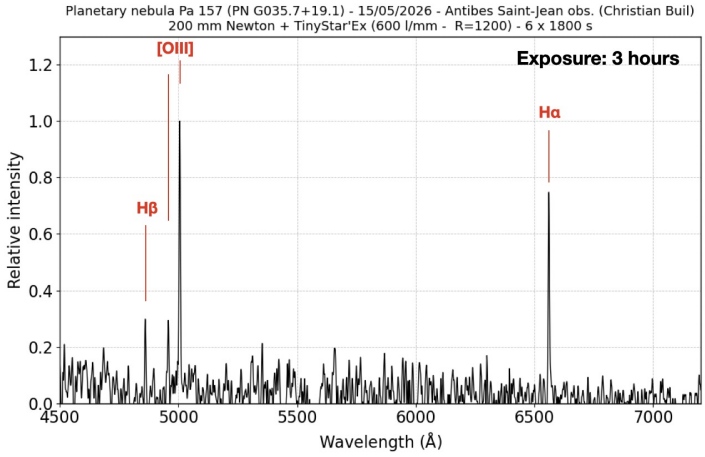
See **Planetary Nebulae.net** website: offers all the useful and practical information about the planetary nebulae (candidates), how to apply for a new discovery and how to confirm the candidates.

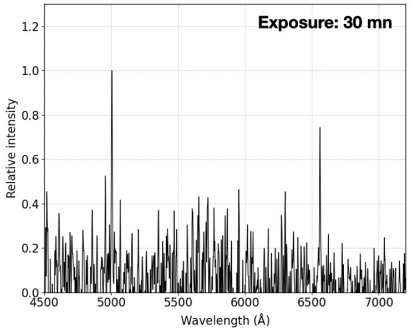


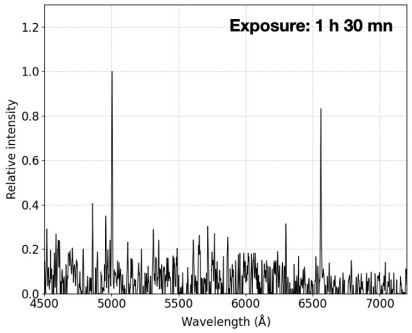
Extracts from the spectrum of the nebula Pa 157 (left: the [O III] and Hβ emission lines; right: the Hα line). The ring structure is clearly detectable, as well as the spectrum of the central star (V = 16.2). Total exposure time: 3 hours. Suburban observing conditions. 200 mm telescope + TinyStar'Ex (at R = 1200 configuration) + Player One Uranus M Pro.



DSS2 images of Pa 157: the nebula is barely detectable
RA: 17:47:09 - DEC: +11:00:21 (2000.0)







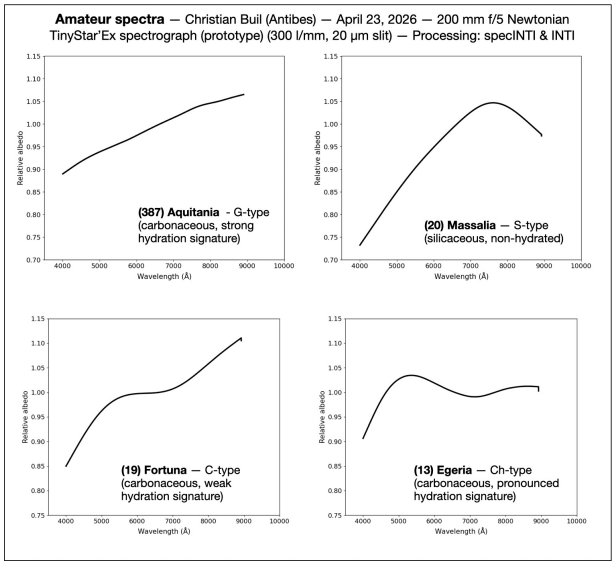
Le spectre des astéroïdes.

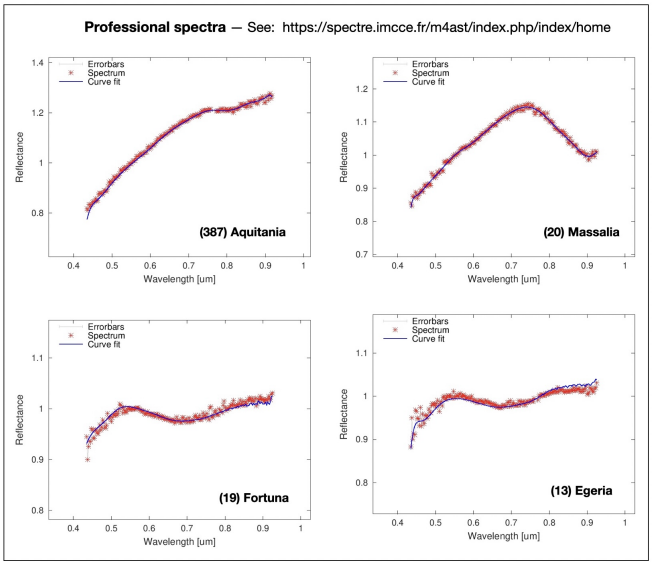
TinyStar'Ex project

Education and Science

Observing asteroids: study of spectral classes (or spectral taxonomy)

The diversity of observed spectral types largely depends on the chemical and mineralogical composition of the materials present on asteroid surfaces. This approach, which relates to the study of their geology, is accessible to amateur astronomers and offers a field of observation that is both original and scientifically rewarding.





Spectroscopie de scènes de tous les jours. Utilisation d'optiques photographiques ou vidéo.



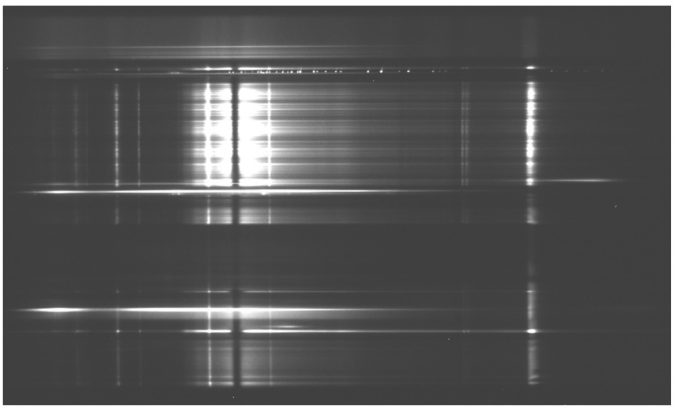
La nature spectrale de l'éclairage de la Tour Eiffet.



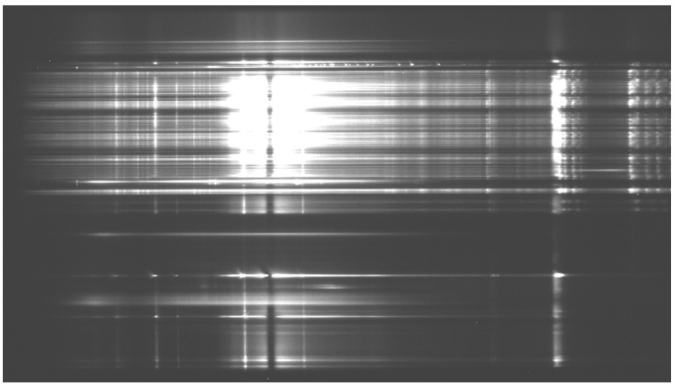
The Eiffel Tower and TinyStar'Ex

F=8.5 mm lens

25° Fied of View

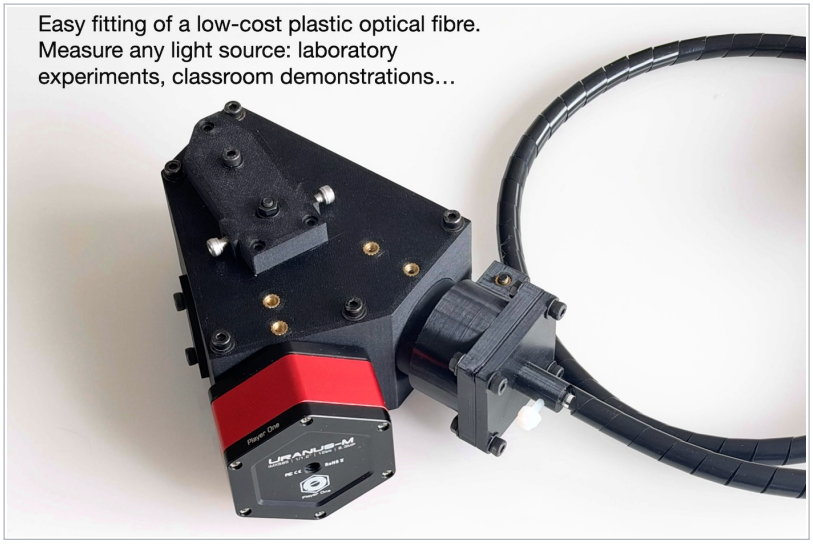


Spectrum of the Eiffel Tower's night lighting

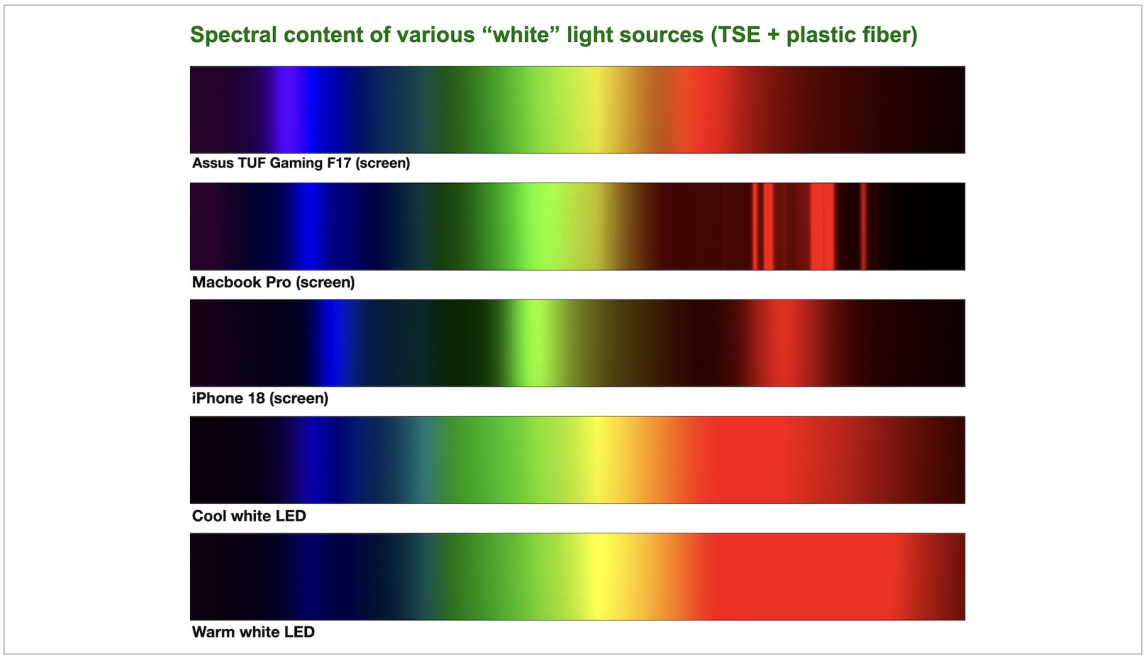


Spectrum of the Eiffel Tower's sparkling lights at the top of the hour

Adaptateur imprimé en 3D permettant d'éclairer la fente d'entrée avec la sortie d'une fibre optique. Il s'agit ici d'un guide de lumière en plastique très économique, de 5 mm de diamètre, vendu pour l'éclairage décoratif et que l'on coupe à la longueur voulue avec une simple paire de ciseaux. Idéal pour des expériences de physique, à l'école ou ailleurs, pour capter la lumière des sources les plus diverses en vue d'une analyse spectrale, ou tout simplement pour se familiariser avec les techniques de la spectroscopie.



Exemple d'utilisation d'une fibre optique : comparaison des spectres produits par des écrans et des lampes à LED. Les images des spectres ont été colorisées a posteriori. Pour cette démonstration, la réponse spectrale de l'instrument n'a pas été corrigée : il s'agit donc de données brutes, telles qu'on les voit en direct, sans traitement.



Observation de cibles très faibles en ville, avec une caméra non refroidie. La spectroscopie compacte, économique et partout!

TinyStar'Ex (TSE)

The spectrum of a magnitude 13.5 supernova (SN2026fvx in NGC 4205) with an uncooled CMOS camera

Cooled camera

Temperature: -20° C

Model: Player One Uranus-M pro
Sony IMX585 sensor

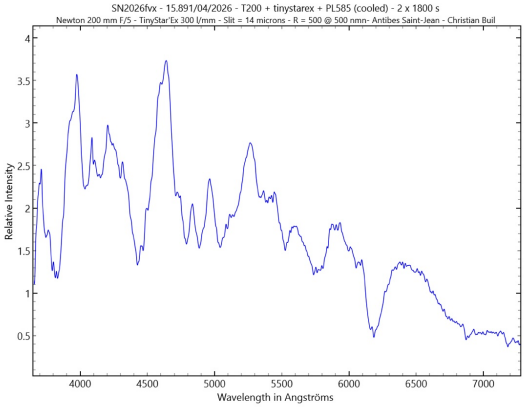
Gain = 220; Offset = 20; Effective gain = 0.911 e^-/ADU; Readout noise = 0.96 e^-; Dark current = 0.00075 e^-/s

TinyStar'Ex spectrograph on 200 mm f/5 Newton telescope, slit GEN2 14 microns, 300 l/mm grating

Total exposure time: 2 x 1800 s (1 hour)
Signal to noise ratio: 150 @ 550 nm

April 15, 2026 - Antibes (France)





Uncooled camera

Temperature: +16° C

Model: Player One Uranus-M
Sony IMX585 sensor

Gain = 220; Offset = 20; Effective gain = 0.911 e^-/ADU; Readout noise = 1.03 e^-; Dark current = 0.226 e^-/s

TinyStar'Ex spectrograph on 200 mm f/5 Newton telescope, slit GEN2 14 microns, 300 l/mm grating

Total exposure time: 20 x 180 s (1 hour)
Signal to noise ratio: 16 @ 550 nm

April 16, 2026 - Antibes (France)

